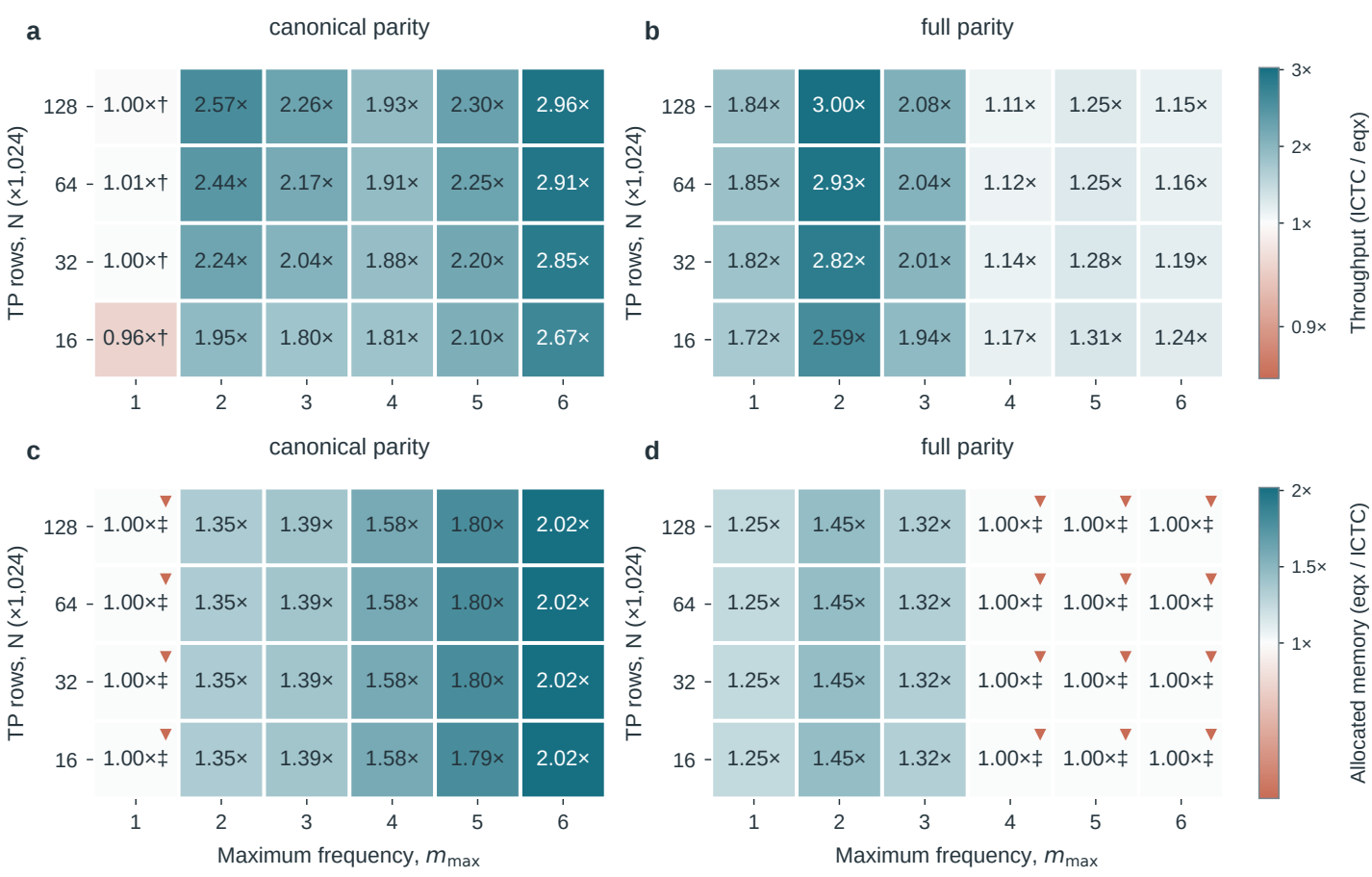

ICTC-O2 HP vs eqx: throughput and peak memory
O(2) tensor products · Forward · C = 64 · N = 16,384–131,072 · m_max = 1–6
a
canonical parity
b
full parity
c
canonical parity
d
full parity
TP rows, N (×1,024)
128
64
32
16
1.00×† 2.57× 2.26× 1.93× 2.30× 2.96×
1.01×† 2.44× 2.17× 1.91× 2.25× 2.91×
1.00×† 2.24× 2.04× 1.88× 2.20× 2.85×
0.96×† 1.95× 1.80× 1.81× 2.10× 2.67×
1.84× 3.00× 2.08× 1.11× 1.25× 1.15×
1.85× 2.93× 2.04× 1.12× 1.25× 1.16×
1.82× 2.82× 2.01× 1.14× 1.28× 1.19×
1.72× 2.59× 1.94× 1.17× 1.31× 1.24×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.79× 2.02×
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1 2 3 4 5 6
Maximum frequency, $m_{\max}$
Throughput (ICTC / eqx)
3×
2×
1×
0.9×
Allocated memory (eqx / ICTC)
2×
1.5×
1×
† IQR overlap * isolated timing ‡ memory increase
v0.1.5